Enduring Terror: The Effects of the Boston Marathon Bombing on Competitive Performance

Alexander Cardazzi Zachary Rodriguez

Virginia Association of Economists
Harrisonburg, VA

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Introduction

Large scale terror attacks, while relatively uncommon, imposes large costs onto both society and individuals.

Terrorist events like the 9/11 Attacks (Metcalfe et al., 2011) and the 2013 Boston Marathon Bombing (Clark et al., 2020) reduced self-reported happiness and increased stress among the American population in the week (or two) following the attack.

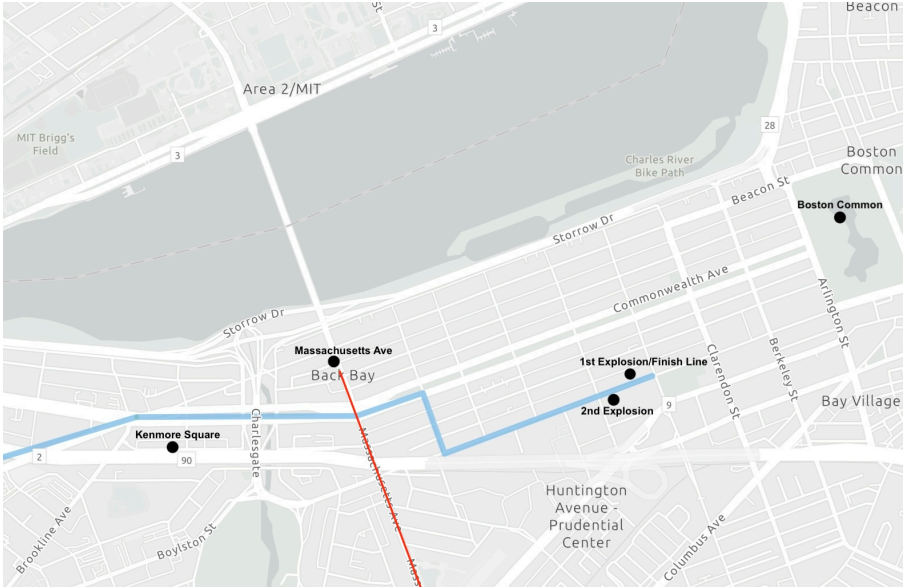
We ask: what are the effects of *direct* experience of a terrorist attack on future performance & participation in competitive environments?

The Boston Marathon

The Boston Marathon is the world's oldest annual marathon (est. 1897) and one of the most prestigious. Only other "World Majors" in the US are in NYC and Chicago.

- Athletes must* qualify for the Boston Marathon.
- Qualifying times are relative to age and gender.
- The race is organized into waves (2013 had three).
- The final wave contains "charity" runners.

On April 15, 2013, at 2:49 PM, two bombs detonated near the finish line, killing three people and wounding hundreds more.


Area 2/MIT
MIT Brigg's Field
Charles River Bike Path
Boston Common
Beacon St
Storrow Dr
Commonwealth Ave
Massachusetts Ave
Back Bay
1st Explosion/Finish Line
2nd Explosion
Kenmore Square
Charlesgate
Clarendon St
Berkeley St
Arlington St
Bay Village
Huntington Avenue - Prudential Center
Columbus Ave
Boylston St
Brookline Ave

The 117th Boston Marathon

Finishing Status by Wave:

- 1.6% of athletes in wave one DNF.
- 4.2% of athletes in wave two DNF.
- 64% of athletes in wave three DNF.
  - 1,183 out of 2,663 (44.4%) non-charity runners DNF.
  - 3,992 out of 5,429 (73.5%) charity runners DNF.

We focus on runners in wave three, and observe their performance across other races.

Data Sources

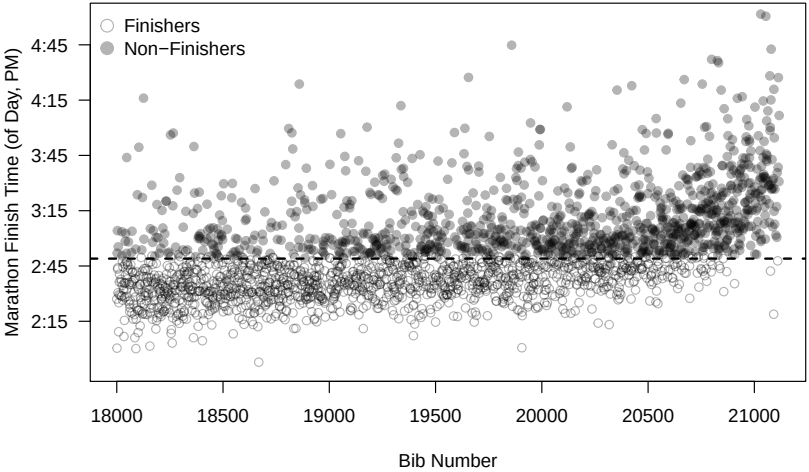
From the Boston Athletic Association, we collect performance "split times," bib numbers, demographics (gender, age, residence) and registration information for the 2009-2017 Marathons (and 2001-2008).

We also collect similar information from the New York City, Chicago, and Marine Corps (Washington DC) Marathons.

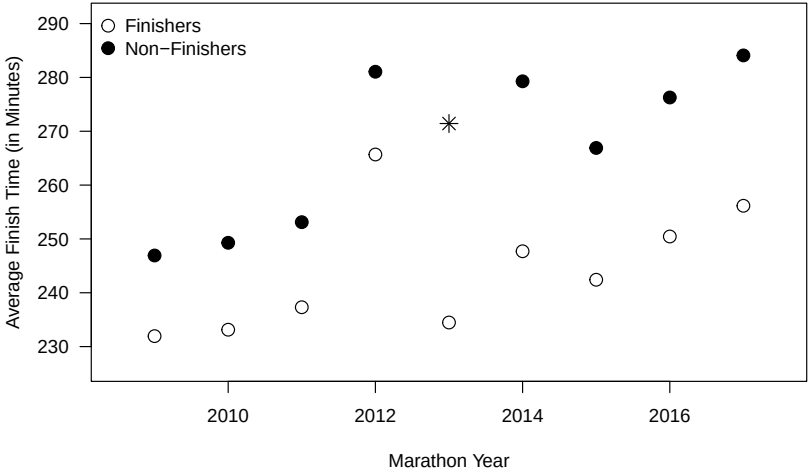
Summary Statistics from 2013

Race Status		Mean	St. Dev.	Min	Median	Max
Finishers	Fin. Time	234.47	9.07	192.72	235.03	255.62
	Qual. Time	238.33	9.87	195	235	295
	Bib Num.	19192.77	769.07	18000	19113	21110
	Male	0.18	0.39	0	0	1
	Age	51.68	7.32	25	50	80
	Foreign	0.2	0.4	0	0	1
Non-Finishers	Fin. Time	271.44	22.65	237.35	264.25	484.98
	Qual. Time	246.47	15.8	185	240	310
	Bib Num.	20008.29	833.91	18002	20185.5	21117
	Male	0.31	0.46	0	0	1
	Age	57.04	8.67	23	57	82
	Foreign	0.2	0.4	0	0	1

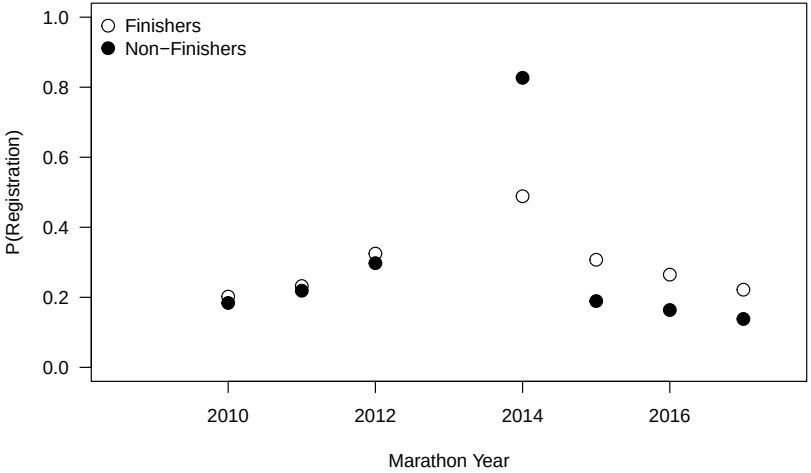
2013 Bib Number vs Finish Time



Performance Over Time



Participation Over Time



Research Design

We model athlete i's performance in marathon t as:

$$\text{Time}_{it} = \delta\ (\text{DNF2013}_i \times \text{Post}_t) + \beta\ \text{Bib}_{it} + \mu_i + \tau_{a(i)g(i)t} + \epsilon_{it} \tag{1}$$

where:

δ measures the relative performance change for diverted runners compared to those who finished

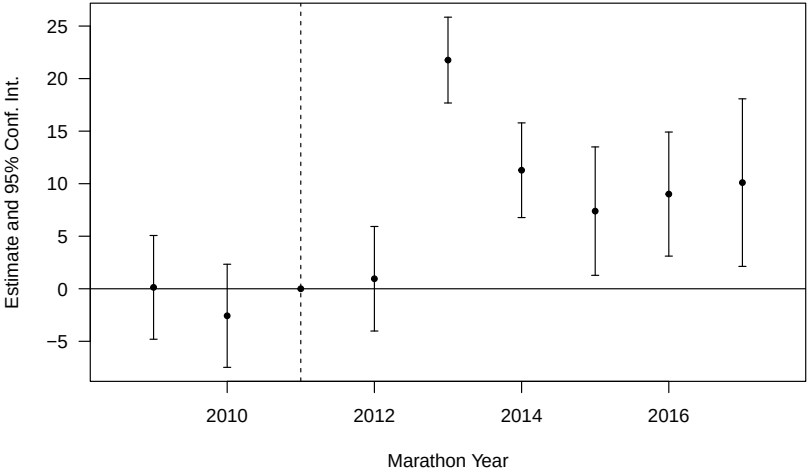
β controls for pre-race expectations

μ_i is a runner-specific fixed effect

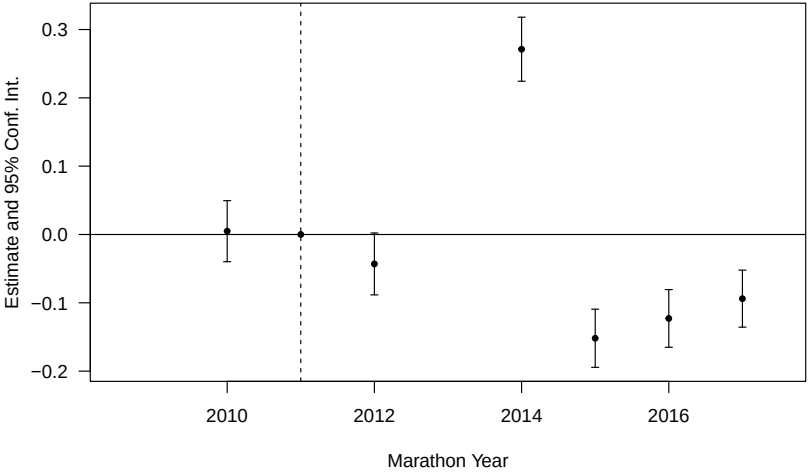
$\tau_{a(i)g(i)t}$ is an age-gender-year fixed effect

Then, we match both finishers and non-finishers to runners absent from the 2013 marathon to 1) estimate treatment effects using an alternative control group and 2) estimate the effects on finishers.

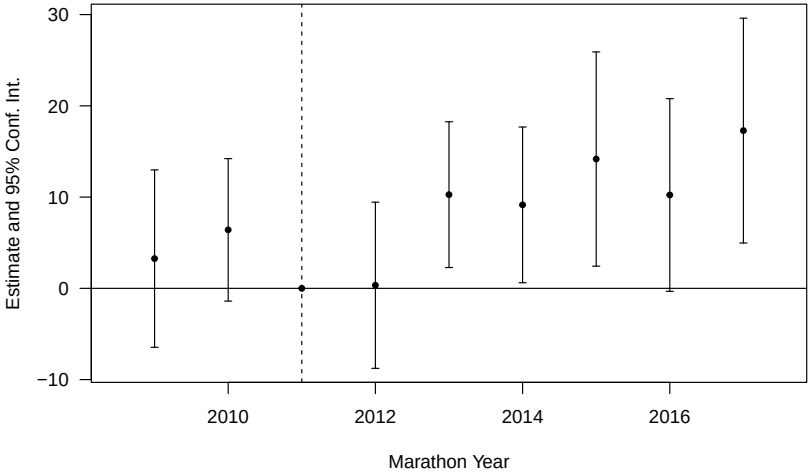
Results: Performance in Boston



Results: Participation in Boston



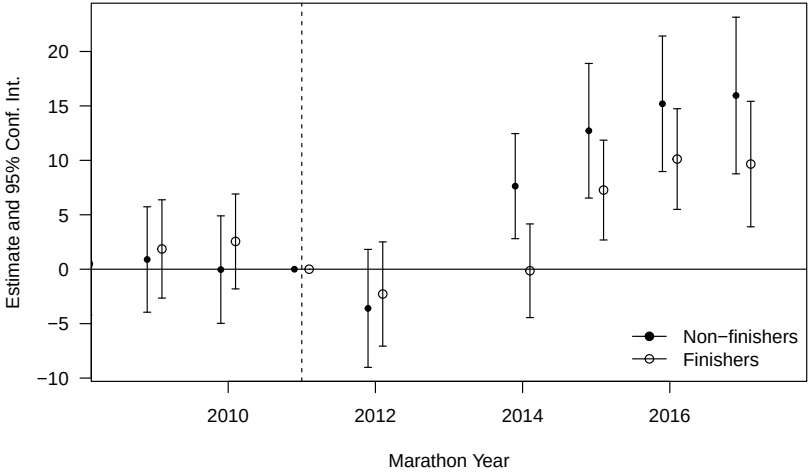
Results: Performance Outside Boston



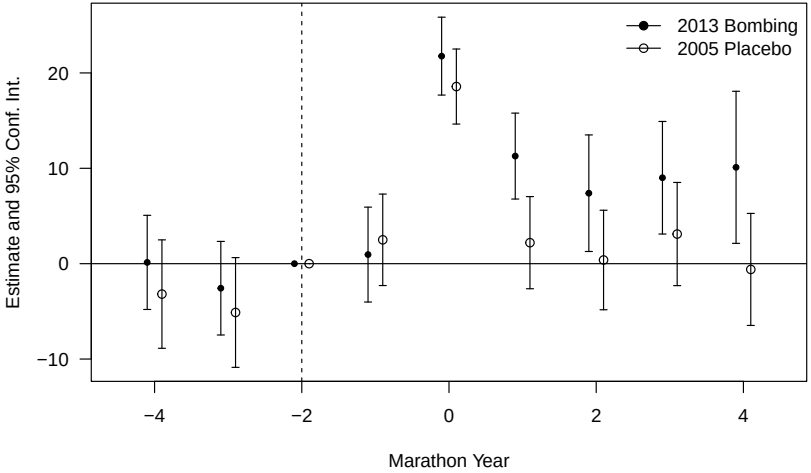
Results: Participation Outside Boston

	Chicago	Washington DC	New York City	All
DNF2013 x Post	-0.0421***	-0.0056	0.0027	-0.0478***
	(0.0115)	(0.0081)	(0.0131)	(0.0163)
R2 Adj.	0.151	0.131	0.127	0.098
Num.Obs.	10576	11898	10576	11898

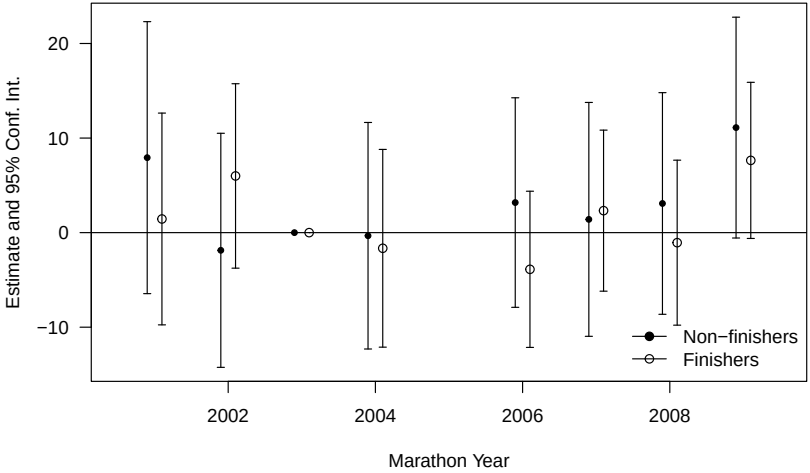
Results: Matched Sample



Results: 2005 "Placebo" Bombing



Results: 2005 "Placebo" Bombing + Matched Sample



Summary

1. The effects of the Boston Marathon Bombing on runners present (both finishers and non-finishers) were long-lasting, unlike the effects on the general population.
2. We find strong performance declines (about 4%) for at least four years. These performance declines also cause reduced participation (runners did not opt out, they were pushed out).
3. As a reference point, in 2017, Nike released the Alphafly shoe which reduced times for elite runners by approximately 4%, and were promptly "outlawed."
4. Our evidence is supported by outcomes in other major marathons, a matching analysis, and a placebo analysis.

Conclusion

- We find terrorist attacks produce sustained reductions in performance even among a highly motivated group of individuals.
- These results have important implications for labor markets: trauma reduces output which could lead to reduced earnings through lower wages (intensive margin) or unemployment (extensive margin).
- Therefore, support for victims must remain in place over long time horizons to be effective (e.g., The Marathon Bombing Victim Assistance program).

References I

Clark, A. E., Doyle, O., and Stancanelli, E. (2020). The Impact of Terrorism on Individual Well-Being: Evidence from the Boston Marathon Bombing. *The Economic Journal*, 130(631):2065–2104.

Metcalfe, R., Powdthavee, N., and Dolan, P. (2011). Destruction and Distress: Using a Quasi-Experiment to Show the Effects of the September 11 Attacks on Mental Well-Being in the United Kingdom. *The Economic Journal*, 121(550):F81–F103.